

$$\begin{aligned}\Pr\{X - np \geq r\} &= \sum_{k=\lceil np+r \rceil}^n b(k; n, p) \\ &\leq \left(\frac{npe}{r}\right)^r.\end{aligned}$$

Proof By equation (C.41), we have $\mu = E[X] = np$. ■

Exercises

★ **C.5-1**

Which is more likely: getting exactly n heads in $2n$ flips of a fair coin, or n heads in n flips of a fair coin?

★ **C.5-2**

Prove Corollaries C.6 and C.7.

★ **C.5-3**

Show that

$$\sum_{i=0}^{k-1} \binom{n}{i} a^i < (a+1)^n \frac{k}{na - k(a+1)} b(k; n, a/(a+1))$$

for all $a > 0$ and all k such that $0 < k < na/(a+1)$.

★ **C.5-4**

Prove that if $0 < k < np$, where $0 < p < 1$ and $q = 1 - p$, then

$$\sum_{i=0}^{k-1} p^i q^{n-i} < \frac{kq}{np - k} \left(\frac{np}{k}\right)^k \left(\frac{nq}{n-k}\right)^{n-k}.$$

★ **C.5-5**

Use Theorem C.8 to show that

$$\Pr\{\mu - X \geq r\} \leq \left(\frac{(n-\mu)e}{r}\right)^r$$

for $r > n - \mu$. Similarly, use Corollary C.9 to show that

$$\Pr\{np - X \geq r\} \leq \left(\frac{nqe}{r}\right)^r$$

for $r > n - np$.

★ C.5-6

Consider a sequence of n Bernoulli trials, where in the i th trial, for $i = 1, 2, \dots, n$, success occurs with probability p_i and failure occurs with probability $q_i = 1 - p_i$. Let X be the random variable describing the total number of successes, and let $\mu = E[X]$. Show that for $r \geq 0$,

$$\Pr\{X - \mu \geq r\} \leq e^{-r^2/2n}.$$

(Hint: Prove that $p_i e^{\alpha q_i} + q_i e^{-\alpha p_i} \leq e^{\alpha^2/2}$. Then follow the outline of the proof of Theorem C.8, using this inequality in place of inequality (C.49).)

★ C.5-7

Show that choosing $\alpha = \ln(r/\mu)$ minimizes the right-hand side of inequality (C.51).

Problems
C-1 The Monty Hall problem

Imagine that you are a contestant in the 1960s game show *Let's Make a Deal*, hosted by emcee Monty Hall. A valuable prize is hidden behind one of three doors and comparatively worthless prizes behind the other two doors. You will win the valuable prize, typically an automobile or other expensive product, if you select the correct door. After you have picked one door, but before the door has been opened, Monty, who knows which door hides the automobile, directs his assistant Carol Merrill to open one of the other doors, revealing a goat (not a valuable prize). He asks whether you would like to stick with your current choice or to switch to the other closed door. What should you do to maximize your chances of winning the automobile and not the other goat?

The answer to this question—stick or switch?—has been heavily debated, in part because the problem setup is ambiguous. We'll explore different subtle assumptions.

- a.** Suppose that your first pick is random, with probability $1/3$ of choosing the right door. Moreover, you know that Monty always gives every contestant (and will give you) the opportunity to switch. Prove that it is better to switch than stick. What is your probability of winning the automobile?

This answer is the one typically given, even though the original statement of the problem rarely mentions the assumption that Monty *always* offers the contestant the opportunity to switch. But, as the remainder of this problem will elucidate, your best strategy may be different if this unstated assumption does not hold. In

fact, in the real game show, after a contestant picked a door, Monty sometimes simply asked Carol to open the door that the contestant had chosen.

Let's model the interactions between you and Monty as a probabilistic experiment, where you both employ randomized strategies. Specifically, after you pick a door, Monty offers you the opportunity to switch with probability p_{right} if you picked the right door and with probability p_{wrong} if you picked the wrong door. Given the opportunity to switch, you randomly choose to switch with probability p_{switch} . For example, if Monty always offers you the opportunity to switch, then his strategy is given by $p_{\text{right}} = p_{\text{wrong}} = 1$. If you always switch, then your strategy is given by $p_{\text{switch}} = 1$.

The game can now be viewed as an experiment consisting of five steps:

1. You pick a door at random, choosing the automobile (right) with probability $1/3$ or a goat (wrong) with probability $2/3$.
2. Carol opens one of the two closed doors, revealing a goat.
3. Monty offers you the opportunity to switch with probability p_{right} if your choice is right and with probability p_{wrong} if your choice is wrong.
4. If Monty makes you an offer in step 3, you switch with probability p_{switch} .
5. Carol opens the door you've chosen, revealing either an automobile (you win) or a goat (you lose).

Let's now analyze this game and understand how the choices of p_{right} , p_{wrong} , and p_{switch} influence the probability of winning.

- b.** What are the six outcomes in the sample space for this game? Which outcomes correspond to you winning the automobile? What are the probabilities in terms of p_{right} , p_{wrong} , and p_{switch} of each outcome? Organize your answers into a table.
- c.** Use the results of your table (or other means) to prove that the probability of winning the automobile is

$$\frac{1}{3}(2p_{\text{wrong}}p_{\text{switch}} - p_{\text{right}}p_{\text{switch}} + 1).$$

Suppose that Monty knows the probability p_{switch} that you switch, and his goal is to minimize your chance of winning.

- d.** If $p_{\text{switch}} > 0$ (you switch with a positive probability), what is Monty's best strategy, that is, his best choice for p_{right} and p_{wrong} ?
- e.** If $p_{\text{switch}} = 0$ (you always stick), argue that all of Monty's possible strategies are optimal for him.

Suppose that now Monty's strategy is fixed, with particular values for p_{right} and p_{wrong} .

- f.* If you know p_{right} and p_{wrong} , what is your best strategy for choosing your probability p_{switch} of switching as a function of p_{right} and p_{wrong} ?
- g.* If you don't know p_{right} and p_{wrong} , what choice of p_{switch} maximizes the minimum probability of winning over all the choices of p_{right} and p_{wrong} ?

Let's return to the original problem as stated, where Monty has given you the option of switching, but you have no knowledge of Monty's possible motivations or strategies.

- h.* Argue that the conditional probability of winning the automobile given that Monty offers you the opportunity to switch is

$$\frac{p_{\text{right}} - p_{\text{right}} p_{\text{switch}} + 2p_{\text{wrong}} p_{\text{switch}}}{p_{\text{right}} + 2p_{\text{wrong}}}. \quad (\text{C.52})$$

Explain why $p_{\text{right}} + 2p_{\text{wrong}} \neq 0$.

- i.* What is the value of expression (C.52) when $p_{\text{switch}} = 1/2$? Show that choosing $p_{\text{switch}} < 1/2$ or $p_{\text{switch}} > 1/2$ allows Monty to select values for p_{right} and p_{wrong} that yield a lower value for expression (C.52) than choosing $p_{\text{switch}} = 1/2$.
- j.* Suppose that you don't know Monty's strategy. Explain why choosing to switch with probability $1/2$ is a good strategy for the original problem as stated. Summarize what you have learned overall from this problem.

C-2 Balls and bins

This problem investigates the effect of various assumptions on the number of ways of placing n balls into b distinct bins.

- a.* Suppose that the n balls are distinct and that their order within a bin does not matter. Argue that the number of ways of placing the balls in the bins is b^n .
- b.* Suppose that the balls are distinct and that the balls in each bin are ordered. Prove that there are exactly $(b + n - 1)!/(b - 1)!$ ways to place the balls in the bins. (*Hint:* Consider the number of ways of arranging n distinct balls and $b - 1$ indistinguishable sticks in a row.)
- c.* Suppose that the balls are identical, and hence their order within a bin does not matter. Show that the number of ways of placing the balls in the bins is $\binom{b+n-1}{n}$. (*Hint:* Of the arrangements in part (b), how many are repeated if the balls are made identical?)

- d.* Suppose that the balls are identical and that no bin may contain more than one ball, so that $n \leq b$. Show that the number of ways of placing the balls is $\binom{b}{n}$.
- e.* Suppose that the balls are identical and that no bin may be left empty. Assuming that $n \geq b$, show that the number of ways of placing the balls is $\binom{n-1}{b-1}$.

Appendix notes

The first general methods for solving probability problems were discussed in a famous correspondence between B. Pascal and P. de Fermat, which began in 1654, and in a book by C. Huygens in 1657. Rigorous probability theory began with the work of J. Bernoulli in 1713 and A. De Moivre in 1730. Further developments of the theory were provided by P.-S. Laplace, S.-D. Poisson, and C. F. Gauss.

Sums of random variables were originally studied by P. L. Chebyshev and A. A. Markov. A. N. Kolmogorov axiomatized probability theory in 1933. Chernoff [91] and Hoeffding [222] provided bounds on the tails of distributions. Seminal work in random combinatorial structures was done by P. Erdős.

Knuth [259] and Liu [302] are good references for elementary combinatorics and counting. Standard textbooks such as Billingsley [56], Chung [93], Drake [125], Feller [139], and Rozanov [390] offer comprehensive introductions to probability.

D Matrices

Matrices arise in numerous applications, including, but by no means limited to, scientific computing. If you have seen matrices before, much of the material in this appendix will be familiar to you, but some of it might be new. Section D.1 covers basic matrix definitions and operations, and Section D.2 presents some basic matrix properties.

D.1 Matrices and matrix operations

This section reviews some basic concepts of matrix theory and some fundamental properties of matrices.

Matrices and vectors

A *matrix* is a rectangular array of numbers. For example,

$$\begin{aligned} A &= \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} \\ &= \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \end{aligned} \tag{D.1}$$

is a 2×3 matrix $A = (a_{ij})$, where for $i = 1, 2$ and $j = 1, 2, 3$, the element of the matrix in row i and column j is denoted by a_{ij} . By convention, uppercase letters denote matrices and corresponding subscripted lowercase letters denote their elements. We denote the set of all $m \times n$ matrices with real-valued entries by $\mathbb{R}^{m \times n}$ and, in general, the set of $m \times n$ matrices with entries drawn from a set S by $S^{m \times n}$.

The *transpose* of a matrix A is the matrix A^T obtained by exchanging the rows and columns of A . For the matrix A of equation (D.1),

$$A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

A **vector** is a one-dimensional array of numbers. For example,

$$x = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$$

is a vector of size 3. We sometimes call a vector of length n an **n -vector**. By convention, lowercase letters denote vectors, and the i th element of a size- n vector x is denoted by x_i , for $i = 1, 2, \dots, n$. We take the standard form of a vector to be as a **column vector** equivalent to an $n \times 1$ matrix, whereas the corresponding **row vector** is obtained by taking the transpose:

$$x^T = (2 \ 3 \ 5).$$

The **unit vector** e_i is the vector whose i th element is 1 and all of whose other elements are 0. Usually, the context makes the size of a unit vector clear.

A **zero matrix** is a matrix all of whose entries are 0. Such a matrix is often denoted 0, since the ambiguity between the number 0 and a matrix of 0s can usually be resolved from context. If a matrix of 0s is intended, then the size of the matrix also needs to be derived from the context.

Square matrices

Square $n \times n$ matrices arise frequently. Several special cases of square matrices are of particular interest:

1. A **diagonal matrix** has $a_{ij} = 0$ whenever $i \neq j$. Because all of the off-diagonal elements are 0, a succinct way to specify the matrix lists only the elements along the diagonal:

$$\text{diag}(a_{11}, a_{22}, \dots, a_{nn}) = \begin{pmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{pmatrix}.$$

2. The $n \times n$ **identity matrix** I_n is a diagonal matrix with 1s along the diagonal:

$$\begin{aligned} I_n &= \text{diag}(1, 1, \dots, 1) \\ &= \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}. \end{aligned}$$

When I appears without a subscript, its size derives from the context. The i th column of an identity matrix is the unit vector e_i .

3. A **tridiagonal matrix** T is one for which $t_{ij} = 0$ if $|i - j| > 1$. Nonzero entries appear only on the main diagonal, immediately above the main diagonal ($t_{i,i+1}$ for $i = 1, 2, \dots, n - 1$), or immediately below the main diagonal ($t_{i+1,i}$ for $i = 1, 2, \dots, n - 1$):

$$T = \begin{pmatrix} t_{11} & t_{12} & 0 & 0 & \dots & 0 & 0 & 0 \\ t_{21} & t_{22} & t_{23} & 0 & \dots & 0 & 0 & 0 \\ 0 & t_{32} & t_{33} & t_{34} & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & t_{n-2,n-2} & t_{n-2,n-1} & 0 \\ 0 & 0 & 0 & 0 & \dots & t_{n-1,n-2} & t_{n-1,n-1} & t_{n-1,n} \\ 0 & 0 & 0 & 0 & \dots & 0 & t_{n,n-1} & t_{nn} \end{pmatrix}.$$

4. An **upper-triangular matrix** U is one for which $u_{ij} = 0$ if $i > j$. All entries below the diagonal are 0:

$$U = \begin{pmatrix} u_{11} & u_{12} & \dots & u_{1n} \\ 0 & u_{22} & \dots & u_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & u_{nn} \end{pmatrix}.$$

An upper-triangular matrix is **unit upper-triangular** if it has all 1s along the diagonal.

5. A **lower-triangular matrix** L is one for which $l_{ij} = 0$ if $i < j$. All entries above the diagonal are 0:

$$L = \begin{pmatrix} l_{11} & 0 & \dots & 0 \\ l_{21} & l_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ l_{n1} & l_{n2} & \dots & l_{nn} \end{pmatrix}.$$

A lower-triangular matrix is **unit lower-triangular** if it has all 1s along the diagonal.

6. A **permutation matrix** P has exactly one 1 in each row or column, and 0s elsewhere. An example of a permutation matrix is

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}.$$

Such a matrix is called a permutation matrix because multiplying a vector x by a permutation matrix has the effect of permuting (rearranging) the elements of x . Exercise D.1-4 explores additional properties of permutation matrices.

7. A **symmetric matrix** A satisfies the condition $A = A^T$. For example,

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 6 & 4 \\ 3 & 4 & 5 \end{pmatrix}$$

is a symmetric matrix.

Basic matrix operations

The elements of a matrix or vector are **scalar numbers** from a number system, such as the real numbers, the complex numbers, or integers modulo a prime. The number system defines how to add and multiply scalars. These definitions extend to encompass addition and multiplication of matrices.

We define **matrix addition** as follows. If $A = (a_{ij})$ and $B = (b_{ij})$ are $m \times n$ matrices, then their matrix sum $C = (c_{ij}) = A + B$ is the $m \times n$ matrix defined by

$$c_{ij} = a_{ij} + b_{ij}$$

for $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$. That is, matrix addition is performed componentwise. A zero matrix is the identity for matrix addition:

$$A + 0 = A = 0 + A.$$

If λ is a scalar number and $A = (a_{ij})$ is a matrix, then $\lambda A = (\lambda a_{ij})$ is the **scalar multiple** of A obtained by multiplying each of its elements by λ . As a special case, we define the **negative** of a matrix $A = (a_{ij})$ to be $-1 \cdot A = -A$, so that the ij th entry of $-A$ is $-a_{ij}$. Thus,

$$A + (-A) = 0 = (-A) + A.$$

The negative of a matrix defines **matrix subtraction**: $A - B = A + (-B)$.

We define **matrix multiplication** as follows. Start with two matrices A and B that are **compatible** in the sense that the number of columns of A equals the number of rows of B . (In general, an expression containing a matrix product AB is always assumed to imply that matrices A and B are compatible.) If $A = (a_{ik})$ is a $p \times q$ matrix and $B = (b_{kj})$ is a $q \times r$ matrix, then their matrix product $C = AB$ is the $p \times r$ matrix $C = (c_{ij})$, where

$$c_{ij} = \sum_{k=1}^q a_{ik} b_{kj} \quad (\text{D.2})$$

for $i = 1, 2, \dots, p$ and $j = 1, 2, \dots, r$. The procedure RECTANGULAR-MATRIX-MULTIPLY on page 374 implements matrix multiplication in the straightforward manner based on equation (D.2), assuming that C is initialized to 0, using pqr multiplications and $p(q-1)r$ additions for a running time of $\Theta(pqr)$. If the matrices are $n \times n$ square matrices, so that $n = p = q = r$, the pseudocode reduces to MATRIX-MULTIPLY on page 81, whose running time is $\Theta(n^3)$. (Section 4.2 describes an asymptotically faster $\Theta(n^{\lg 7})$ -time algorithm due to V. Strassen.)

Matrices have many (but not all) of the algebraic properties typical of numbers. Identity matrices are identities for matrix multiplication:

$$I_m A = A I_n = A$$

for any $m \times n$ matrix A . Multiplying by a zero matrix gives a zero matrix:

$$A \cdot 0 = 0.$$

Matrix multiplication is associative:

$$A(BC) = (AB)C$$

for compatible matrices A , B , and C . Matrix multiplication distributes over addition:

$$\begin{aligned} A(B + C) &= AB + AC, \\ (B + C)D &= BD + CD. \end{aligned}$$

For $n > 1$, multiplication of $n \times n$ matrices is not commutative. For example, if $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, then $AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $BA = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$.

We define matrix-vector products or vector-vector products as if the vector were the equivalent $n \times 1$ matrix (or a $1 \times n$ matrix, in the case of a row vector). Thus, if A is an $m \times n$ matrix and x is an n -vector, then Ax is an m -vector. If x and y are n -vectors, then